

MT PLEASANT, MI, USA

WMO: 725424

Lat: 43.622N

Lon: 84.737W

Elev: 230

StdP: 98.59

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 04885

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.8	-22.2	0.5	-16.9	-19.7	0.7	-14.3	9.4	-4.1	8.5	-3.1	1.5	240	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.1	22.3	30.1	21.2	28.4	20.3	24.1	29.0	23.0	27.5	22.2	26.4	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.9	26.1	22.0	17.1	25.4	21.0	16.1	24.6	73.7	29.1	69.5	27.1	66.0	25.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.7	DB	-22.0	33.9	3.6	1.5	-24.6	35.0	-26.7	35.9	-28.8	36.7	-31.4	37.8	(4)
(5)			WB	-22.3	25.7	3.6	1.3	-24.9	26.7	-27.0	27.5	-29.0	28.2	-31.6	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-5.1	-4.7	0.7	7.5	13.8	18.9	21.4	20.4	16.6	9.9	3.7	-1.7	(6)	
	DBStd	10.51	5.84	5.55	6.10	5.06	4.79	3.68	3.24	3.03	4.18	4.71	4.68	4.78	(7)	
	HDD10.0	1921	468	411	296	106	16	0	0	0	2	62	196	364	(8)	
	HDD18.3	3867	726	644	547	326	156	36	8	14	82	265	440	623	(9)	
	CDD10.0	1383	0	0	7	32	134	268	354	323	199	59	7	1	(10)	
	CDD18.3	286	0	0	1	2	17	53	104	78	28	3	0	0	(11)	
	CDH23.3	2878	0	0	7	30	199	561	1051	707	286	38	0	0	(12)	
	CDH26.7	811	0	0	1	4	49	159	340	185	68	5	0	0	(13)	
(14) Wind	WSAvg	2.7	3.0	3.1	3.0	3.3	2.8	2.4	2.1	2.0	2.1	2.7	2.9	3.0	(14)	
(15) Precipitation	PrecAvg	780	39	28	51	79	75	87	71	89	86	68	69	45	(15)	
	PrecMax	1115	94	67	135	180	180	182	154	258	392	181	135	123	(16)	
	PrecMin	524	1	1	2	18	3	10	18	17	1	4	16	5	(17)	
	PrecStd	120	23	18	27	38	43	44	37	51	64	37	30	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	13.6	22.8	27.1	30.7	32.7	33.9	32.7	32.2	27.3	20.0	13.8	(19)	
		MCWB	9.2	10.2	15.5	16.2	20.4	22.6	23.1	22.7	21.6	20.1	14.2	11.6	(20)	
	2%	DB	7.2	7.6	17.1	22.9	27.8	30.7	32.3	30.5	28.5	23.8	15.9	8.9	(21)	
		MCWB	5.3	5.3	12.0	14.1	18.6	20.8	22.9	21.4	20.2	17.0	11.5	7.1	(22)	
	5%	DB	4.6	4.8	12.8	20.6	25.7	28.2	30.2	28.7	26.8	20.9	13.0	6.4	(23)	
		MCWB	3.1	2.6	9.1	12.6	17.3	20.1	21.7	20.6	19.3	15.8	9.6	4.8	(24)	
	10%	DB	2.5	2.6	9.7	17.1	22.9	26.8	28.0	27.3	24.8	17.9	11.2	4.2	(25)	
		MCWB	1.4	0.9	6.6	10.5	16.1	19.2	20.5	19.9	18.5	14.4	8.5	2.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	10.0	16.7	18.1	22.0	24.5	26.0	25.1	23.6	21.3	14.9	12.3	(27)	
		MCDB	10.2	13.5	23.1	24.2	27.6	30.0	30.5	29.2	28.2	25.2	17.6	13.8	(28)	
	2%	WB	5.3	5.7	13.0	15.5	20.5	22.9	24.5	23.6	22.3	19.3	12.5	7.7	(29)	
		MCDB	6.9	7.4	15.5	20.9	25.3	27.5	29.5	27.7	25.9	22.0	15.1	8.4	(30)	
	5%	WB	3.0	2.7	9.8	13.5	19.0	21.8	23.4	22.5	21.1	17.0	10.5	5.1	(31)	
		MCDB	4.2	4.4	12.3	18.8	23.4	26.4	28.1	26.4	24.2	19.8	12.5	6.7	(32)	
	10%	WB	1.5	1.0	6.3	11.4	17.4	20.6	22.3	21.4	19.8	14.7	8.4	2.9	(33)	
		MCDB	2.5	2.4	9.0	16.5	21.2	24.6	26.6	25.1	23.6	17.6	11.0	4.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	8.2	9.8	11.4	12.2	12.0	12.1	12.0	12.2	10.3	8.1	6.0	(35)	
		MCDBR	8.6	9.8	14.7	16.5	15.1	13.8	13.8	13.5	14.4	14.6	11.7	8.5	(36)	
	5% WB	MCWBR	7.4	7.7	9.9	8.7	7.3	6.3	6.0	5.9	6.9	8.4	8.4	7.5	(37)	
		MCWBR	8.1	9.0	13.5	14.5	12.9	11.3	11.4	11.1	11.1	12.2	10.9	8.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.321	0.329	0.364	0.391	0.416	0.407	0.399	0.376	0.336	0.305	0.282	(40)	
		taud	2.356	2.251	2.380	2.318	2.294	2.257	2.286	2.306	2.368	2.478	2.506	2.429	(41)	
	Ebn at Noon		842	864	911	900	881	857	860	855	849	845	823	819	(42)	
		Edh at Noon	79	105	105	122	128	134	129	122	106	84	69	67	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.36	2.25	3.60	4.56	5.38	6.03	6.10	5.18	4.15	2.58	1.54	1.08	(44)	
	RadStd		0.12	0.26	0.40	0.40	0.51	0.36	0.32	0.26	0.39	0.28	0.20	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page